

1. Print a Hollow Diamond Pattern

Problem Statement

A Hollow Diamond Pattern consists of stars (*) forming the outline of a diamond shape, while the inner area remains empty.

Given an integer N, print a hollow diamond having N rows in the upper half and N rows in the lower half.

Example for N = 5:

```
*
 *
*  *
*  *
*   *
*   *
*   *
*   *
*   *
 *
```

Write a Python program to generate the Hollow Diamond Pattern.

Input Format

n

Output Format

Hollow Diamond Pattern

Constraints

$2 \leq n \leq 20$

Test Case 1

Input

4

Output

```
*
 *
*  *
*  *
*   *
*   *
*   *
*   *
 *
```

Test Case 2

Input

3

Output

```
*
 *
*  *
*  *
*   *
 *
```

2. Generate the Power Set of a Given Set

Problem Statement

The Power Set of a set is the collection of all possible subsets, including:

- Empty Set
- Single Element Sets
- Multiple Element Sets
- Original Set

Example:

Input:

{1, 2, 3}

Output:

{}

{1}

{2}

{3}

{1,2}

{1,3}

{2,3}

{1,2,3}

Write a Python program to generate the Power Set of a given set.

Input Format

n
set elements

Output Format

All Possible Subsets

Constraints

$1 \leq n \leq 15$

Test Case 1

Input

3
1 2 3

Output

{}
{1}
{2}
{3}
{1, 2}
{1, 3}
{2, 3}
{1, 2, 3}

Test Case 2

Input

2
10 20

Output

```
{}  
{10}  
{20}  
{10, 20}
```

3. Find Common Elements Among Multiple Sets

Problem Statement

Given multiple sets of integers, find all elements that are present in every set.

The output should contain only the common elements sorted in ascending order.

Example:

Set 1 = {1, 2, 3, 4}

Set 2 = {2, 3, 5}

Set 3 = {2, 3, 6}

Output:

2 3

Write a Python program to find the intersection of multiple sets.

Input Format

```
n  
set1 elements  
set2 elements  
set3 elements
```

Output Format

```
Common Elements
```

Constraints

2 <= number of sets <= 10

1 <= elements in each set <= 1000

Test Case 1

Input

```
3  
1 2 3 4  
2 3 5  
2 3 6
```

Output

```
2 3
```

Test Case 2

Input

```
3  
10 20 30 40  
20 30 50  
20 30 60
```

Output

```
20 30
```